BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Midterm | Semester: Spring 2024
Duration: 1 hour 30 min | Full Marks: 30

CSE 320: Data Communications

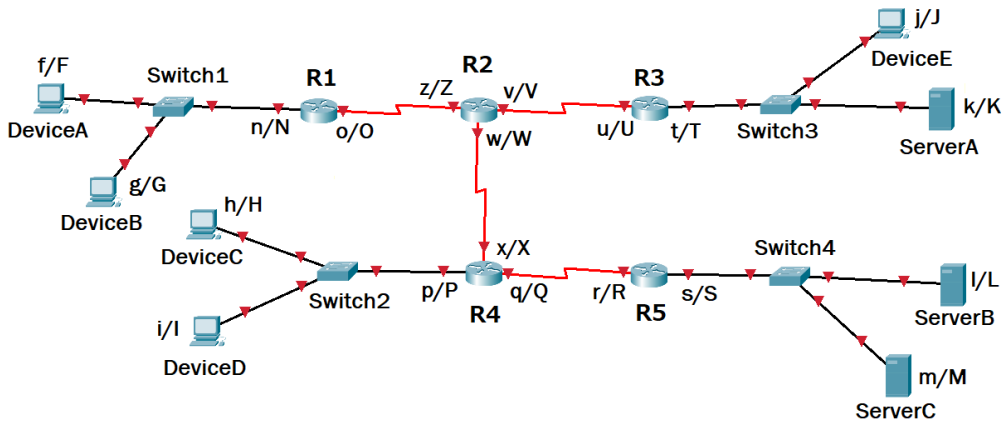
Answer the following questions.
Figures in the right margin indicate marks.

SET B

Name:	ID:	Section:

1. [CO1] a) [2+4]

You have been tasked to work with the network topology shown below. Assume the physical addresses to be the uppercase letters and the logical addresses to be the lowercase letters. For the Devices, use a Port number from the dynamic range (49152 - 65535).

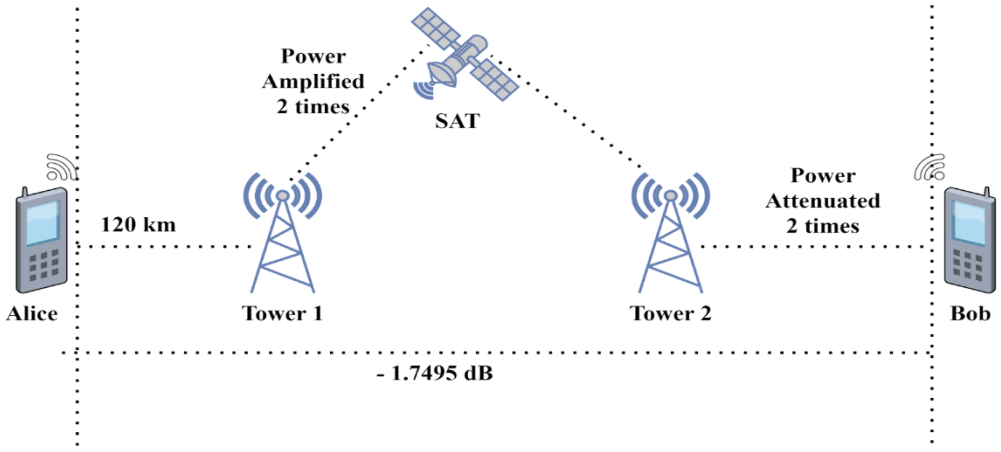


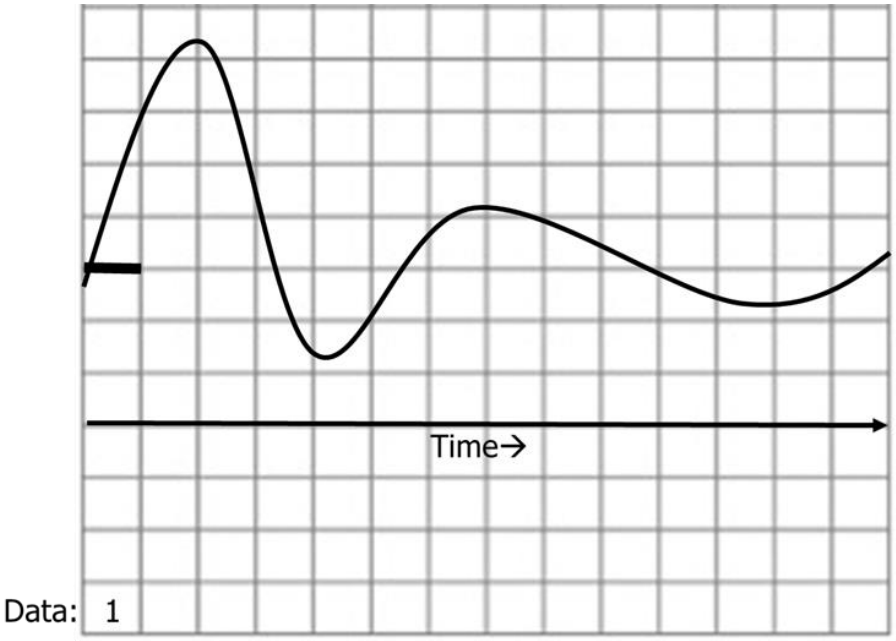
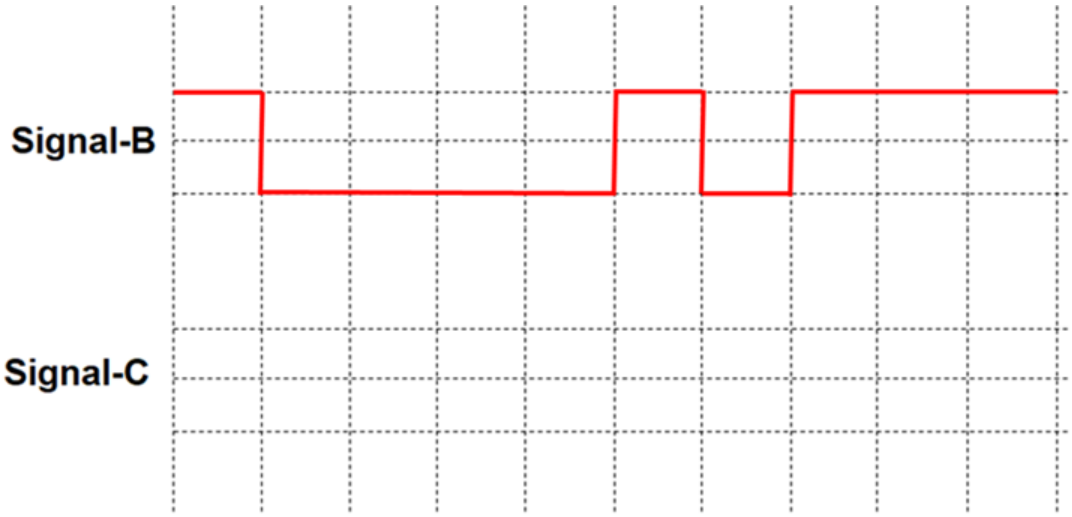
i. How many networks are there in total? What is a process-to-process delivery and which layer of the TCP/IP model is responsible for maintaining this?

ii. Device B is trying to send a data frame to Server C, which is an Email Server using Port 25.

How many hops will there be? **Complete** Frame 1 below by writing the Destination and Source physical, logical and appropriate port addresses if the data frame is in its second last hop.

Frame1	D. Mac	S. Mac	D. IP	S. IP	D. Port	S. Port	Data	Trailer

	b)	Suppose, a hospital has four buildings. Each building has an administration ward and a patient care ward. There are 3 nodes in each ward. Identify the topologies and draw the hybrid network **that satisfy the following criteria:** i) Each ward can communicate with each other as they are connected centrally. If one ward gets disconnected, the other ward remains active. ii) The devices in both wards use a central backbone line and are connected using multipoint connections only. iii) Dedicated point to point connection on both sides is used to connect buildings with each other.	[4]
2. [CO2]	a)	A document carrying 3*10^7 characters take 2 minutes to download over a communication channel, assuming the channel is noisy. The bandwidth of the channel is 150 kHz, analyze the following aspects i. **Calculate** the data rate of the channel. ii. **Determine** the value of the signal to noise ratio (SNR).	[2+2]
	b)	Alice is calling Bob through a hybrid communication system as shown in the following figure. Suppose the signal power is 600 mW at Alice's phone. The power loss rate from Alice to Tower 1 is 1 mW/km. The final attenuation from Alice to Bob is -1.7495 dB. ● **Calculate** the attenuation for the signal transmitted from SAT to Tower 2. 	[3]
	c)	A non-periodic composite signal is characterized by a bandwidth of 15 kHz and a maximum frequency of 25 kHz. The signal exhibits peak amplitudes of 8 V for the lowest and highest frequencies and 14 V for 18 kHz frequency. If the amplitude changes gradually from minimum frequency to maximum frequency and maximum frequency to minimum frequency. **Calculate** the minimum frequency. **Draw** the frequency domain plot.	[3]
3. [CO2]	a)	How to reduce the normalized error in PCM? **Mention** one solution.	[2]

	b)	**Show** the staircase in the following graph and **generate** the digital data from the given analog signal using the Delta Modulation (DM) technique. *You must answer this question in the question paper only.*  	[3]
	c)	Manchester and Differential Manchester solves the consecutive 0's and 1's problem. However, it requires more bandwidth to communicate - **explain** why?	[2]
	d)	 i. Decode the Signal B and **generate** the binary bit stream using NRZ-I line coding scheme. ii. Now from the extracted binary bits, **draw** the Signal-C using an appropriate line coding scheme so that there is no synchronization problem. (draw it in the question paper only)	[3]